

1. Two particles P and Q have masses m and $4m$ respectively. The particles are at rest on a smooth horizontal plane. Particle P is given a horizontal impulse, of magnitude I , in the direction PQ . Particle P then collides directly with Q . Immediately after this collision, P is at rest and Q has speed w . The coefficient of restitution between the particles is e .

(a) Find I in terms of m and w .

(2)

(b) Show that $e = \frac{1}{4}$

(1)

(c) Find, in terms of m and w , the total kinetic energy lost in the collision between P and Q .

(2)

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1. Two particles, P and Q , of masses $3m$ and $2m$ respectively, are moving on a smooth horizontal plane. They are moving in opposite directions along the same straight line when they collide directly.

Immediately before the collision, P is moving with speed $2u$.

The magnitude of the impulse exerted on P by Q in the collision is $\frac{9mu}{2}$

- (a) Find the speed of P immediately after the collision.

(3)

The coefficient of restitution between P and Q is e .

Given that the speed of Q immediately before the collision is u ,

- (b) find the value of e .

(5)

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1. A particle A has mass $2m$ and a particle B has mass $3m$. The particles are moving in opposite directions along the same straight line and collide directly.

Immediately **before** the collision, the speed of A is $2u$ and the speed of B is u .
Immediately **after** the collision, the speed of A is $0.5u$ and the speed of B is w .

Given that the direction of motion of each particle is reversed by the collision,

- (a) find w in terms of u (3)
- (b) find the coefficient of restitution between the particles, (3)
- (c) find, in terms of m and u , the magnitude of the impulse received by A in the collision. (3)

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Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Answer ALL questions. Write your answers in the spaces provided.

1. A small ball of mass 0.3 kg is released from rest from a point 3.6 m above horizontal ground. The ball falls freely under gravity, hits the ground and rebounds vertically upwards.

In the first impact with the ground, the ball receives an impulse of magnitude 4.2 N s . The ball is modelled as a particle.

- (a) Find the speed of the ball immediately after it first hits the ground.

(5)

- (b) Find the kinetic energy lost by the ball as a result of the impact with the ground.

(3)

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2.

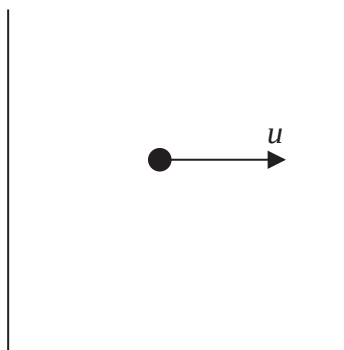


Figure 2

A particle of mass em is at rest on a smooth horizontal plane between two smooth fixed parallel vertical walls, as shown in the plan view in Figure 2. The particle is projected along the plane with speed u towards one of the walls and strikes the wall at right angles. The coefficient of restitution between the particle and each wall is e and air resistance is modelled as being negligible.

Using the model,

- (a) find, in terms of m , u and e , an expression for the total loss in the kinetic energy of the particle as a result of the first two impacts.

Given that e can vary such that $0 < e < 1$ and using the model,

- (b) find the value of e for which the total loss in the kinetic energy of the particle as a result of the first two impacts is a maximum, (4)
- (c) describe the subsequent motion of the particle. (2)



2. Two particles, A and B , of masses $2m$ and $3m$ respectively, are moving on a smooth horizontal plane. The particles are moving in opposite directions towards each other along the same straight line when they collide directly. Immediately before the collision the speed of A is $2u$ and the speed of B is u . In the collision the impulse of A on B has magnitude $5mu$.

(a) Find the coefficient of restitution between A and B .

(9)

(b) Find the total loss in kinetic energy due to the collision.

(4)

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4.

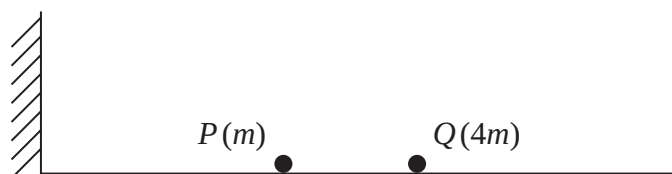


Figure 2

A particle P of mass m and a particle Q of mass $4m$ are at rest on a smooth horizontal plane, as shown in Figure 2.

Particle P is projected with speed u along the plane towards Q and the particles collide.

The coefficient of restitution between the particles is e , where $e > \frac{1}{4}$

As a result of the collision, the direction of motion of P is reversed and P has speed $\frac{u}{5}(4e - 1)$.

- (a) Find, in terms of u and e , the speed of Q after the collision. (3)

After the collision, P goes on to hit a vertical wall which is fixed at right angles to the direction of motion of P .

The coefficient of restitution between P and the wall is f , where $f > 0$

Given that $e = \frac{3}{4}$

- (b) find, in terms of m , u and f , the kinetic energy lost by P as a result of its impact with the wall. Give your answer in its simplest form. (4)

After its impact with the wall, P goes on to collide with Q again.

- (c) Find the complete range of possible values of f . (4)



3. Three particles A , B and C are at rest on a smooth horizontal plane. The particles lie along a straight line with B between A and C .

Particle B has mass $4m$ and particle C has mass km , where k is a positive constant. Particle B is projected with speed u along the plane towards C and they collide directly.

The coefficient of restitution between B and C is $\frac{1}{4}$

- (a) Find the range of values of k for which there would be no further collisions.

(8)

The magnitude of the impulse on B in the collision between B and C is $3mu$

- (b) Find the value of k .

(4)

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4. A small ball, of mass m , is thrown vertically upwards with speed $\sqrt{8gH}$ from a point O on a smooth horizontal floor. The ball moves towards a smooth horizontal ceiling that is a vertical distance H above O . The coefficient of restitution between the ball and the ceiling is $\frac{1}{2}$

In a model of the motion of the ball, it is assumed that the ball, as it moves up or down, is subject to air resistance of constant magnitude $\frac{1}{2}mg$.

Using this model,

- (a) use the work-energy principle to find, in terms of g and H , the speed of the ball immediately before it strikes the ceiling, (5)
- (b) find, in terms of g and H , the speed of the ball immediately before it strikes the floor at O for the first time. (5)

In a simplified model of the motion of the ball, it is assumed that the ball, as it moves up or down, is subject to no air resistance.

Using this simplified model,

- (c) explain, without any detailed calculation, why the speed of the ball, immediately before it strikes the floor at O for the first time, would still be less than $\sqrt{8gH}$ (1)



4. Three particles, P , Q and R , are at rest on a smooth horizontal plane. The particles lie along a straight line with Q between P and R . The particles Q and R have masses m and km respectively, where k is a constant.

Particle Q is projected towards R with speed u and the particles collide directly.

The coefficient of restitution between each pair of particles is e .

- (a) Find, in terms of e , the range of values of k for which there is a second collision. (9)

Given that the mass of P is km and that there is a second collision,

- (b) write down, in terms of u , k and e , the speed of Q after this second collision. (1)

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4.



Figure 1

Three particles, P , Q and R , lie at rest on a smooth horizontal plane. The particles are in a straight line with Q between P and R , as shown in Figure 1.

Particle P is projected **towards** Q with speed u . At the same time, R is projected with speed $\frac{1}{2}u$ **away from** Q , in the direction QR .

Particle P has mass m and particle Q has mass $2m$.

The coefficient of restitution between P and Q is e .

(a) Show that the speed of Q immediately after the collision between P and Q is

$$\frac{u(1+e)}{3} \quad (6)$$

It is given that $e > \frac{1}{2}$

(b) Determine whether there is a collision between Q and R .

(2)

(c) Determine the direction of motion of P immediately after the collision between P and Q .

(2)

(d) Find, in terms of m , u and e , the total kinetic energy lost in the collision between P and Q , simplifying your answer.

(3)

(e) Explain how using $e = 1$ could be used to check your answer to part (d).

(1)

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4. A particle P of mass $3m$ is moving in a straight line on a smooth horizontal floor. A particle Q of mass $5m$ is moving in the opposite direction to P along the same straight line.

The particles collide directly.

Immediately before the collision, the speed of P is $2u$ and the speed of Q is u .

The coefficient of restitution between P and Q is e .

- (a) Show that the speed of Q immediately after the collision is $\frac{u}{8}(9e + 1)$
- (b) Find the range of values of e for which the direction of motion of P is not changed as a result of the collision.

- (b) Find the range of values of e for which the direction of motion of P is not changed as a result of the collision.

When P and Q collide they are at a distance d from a smooth fixed vertical wall, which is perpendicular to their direction of motion. After the collision with P , particle Q collides directly with the wall and rebounds so that there is a second collision between P and Q . This second collision takes place at a distance x from the wall.

Given that $e = \frac{1}{18}$ and the coefficient of restitution between Q and the wall is $\frac{1}{3}$

- (c) find x in terms of d . (6)



- (d) find the full range of possible values of e . (5)